INCH-POUND

MIL-C-11272/3C
7 February 2003
SUPERSEDING
MIL-C-11272/3B
7 January 1970

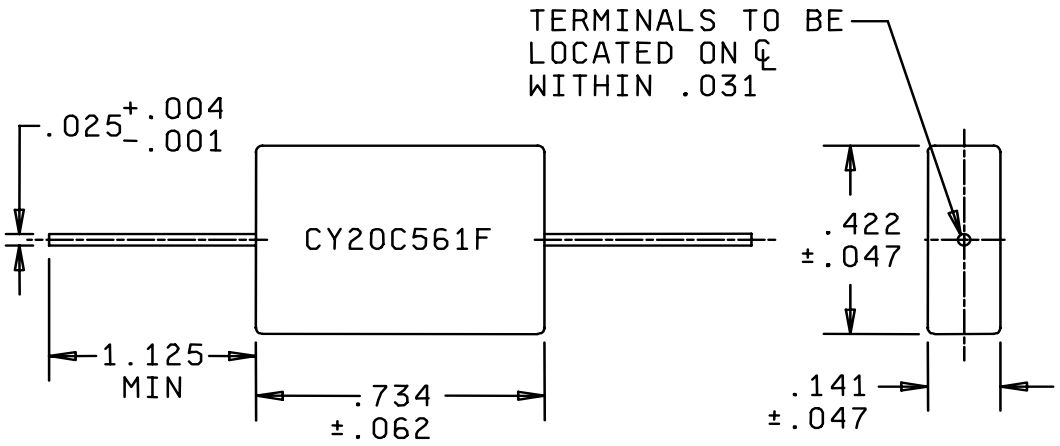
MILITARY SPECIFICATION

CAPACITORS, FIXED, GLASS DIELECTRIC

STYLE CY20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR DESIGN AFTER 7 JAN 1970. USE MIL-PRF-23269.



Inches	mm
.001	.03
.004	.10
.025	.64
.031	.79
.047	1.19
.062	1.57
.141	3.58
.422	10.72
.734	18.64
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

Style CY20 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Requirements: Requirements shall be in accordance with MIL-C-11272, and as specified herein.

Design and construction: Capacitors shall be of the design, construction, and physical dimensions specified on figure 1 and in table I.

TABLE I. Style CY20.

Pin 1/	Capacitance	DC rated voltage	Capacitance tolerances available
	pF	volts	
CY20C561-	560	500	FGJ
CY20C621-	620	500	FGJ
CY20C681-	680	500	FGJ
CY20C751-	750	500	FGJ
CY20C821-	820	500	FGJ
CY20C911-	910	500	FGJ
CY20C102-	1,000	500	FGJ
CY20C112-	1,100	500	FGJ
CY20C122-	1,200	500	FGJ
CY20C132-	1,300	500	FGJ
CY20C152-	1,500	500	FGJ
CY20C162-	1,600	500	FGJ
CY20C182-	1,800	500	FGJ
CY20C202-	2,000	500	FGJ
CY20C222-	2,200	500	FGJ
CY20C242-	2,400	500	FGJ
CY20C272-	2,700	500	FGJ
CY20C302-	3,000	500	FGJ
CY20C332-	3,300	500	FGJ
CY20C362-	3,600	300	FGJ
CY20C392-	3,900	300	FGJ
CY20C432-	4,300	300	FGJ
CY20C472-	4,700	300	FGJ
CY20C512-	5,100	300	FGJ

1/ Complete type designation will include an additional letter symbol to indicate capacitance tolerance, where applicable.

Capacitance: See table I.

Capacitance tolerance: C: ± 0.25 pF, D: ± 0.50 pF, F: 1%, G: 2%, J: 5%.

DC rated voltage: 300 and 500 volts.

Temperature range: -55°C to +125°C.

Life: 150 percent dc rated voltage at 125°C +4°C, -0°C for 2,000 hours.

Group A test: 250 hours: group B continuation test for 1,750 hours.

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Qualification inspection: Capacitors shall meet the following requirements, as applicable:

For qualification test:

At 25°C:

Insulation resistance: 100,000 megohms minimum.

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 125°C value.

Dissipation factor: Not to exceed 0.0025.

At 125°C:

Insulation resistance: Not less than 10,000 megohms.

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 125°C value.

Dissipation factor: Not to exceed 0.007.

Performance check: Capacitors shall meet the following requirements:

For performance check:

Insulation resistance: 100,000 megohms minimum.

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 125°C value.

Dissipation factor: Not to exceed 0.0025.

Temperature coefficient and capacitance drift: The temperature coefficient and capacitance drift shall be within the limits specified in table II.

TABLE II. Temperature coefficient and capacitance drift.

Temperature coefficient	Capacitance drift (-55° to +125°C)
parts/million/°C 140 ±25	0.1 percent or 0.1 pF, whichever is greater.

VERIFICATION:

Sampling and inspection shall be in accordance with MIL-C-11272.

Group A inspection: See MIL-C-11272: Successful performance of group A inspection.

Group B inspection: See MIL-C-11272: Successful performance of group B inspection.

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Qualification: With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, P. O. Box 3990, Columbus OH 43216-5000.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Air Force - 99
DLA - IS

Preparing activity:
DLA - CC

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